

Practice exam 2

Units 3 & 4 Paper 1

Perusal time: 5 minutes

Working time: 90 minutes

Section 1: Multiple choice

Instructions

- Choose the best answer for Questions 1–20.
- This section has **20 questions** and is worth **20 marks**.
- Use a 2B pencil to fill in the A, B, C or D answer bubble completely.
- If you change your mind or make a mistake, use an eraser to remove your response and fill in the new answer bubble completely.

	A	B	C	D
1.	☐	☐	☐	☐
2.	☐	☐	☐	☐
3.	☐	☐	☐	☐
4.	☐	☐	☐	☐
5.	☐	☐	☐	☐
6.	☐	☐	☐	☐
7.	☐	☐	☐	☐
8.	☐	☐	☐	☐
9.	☐	☐	☐	☐
10.	☐	☐	☐	☐
11.	☐	☐	☐	☐
12.	☐	☐	☐	☐
13.	☐	☐	☐	☐
14.	☐	☐	☐	☐
15.	☐	☐	☐	☐
16.	☐	☐	☐	☐
17.	☐	☐	☐	☐
18.	☐	☐	☐	☐
19.	☐	☐	☐	☐
20.	☐	☐	☐	☐

QUESTION 1

Deduce the oxidation state of nitrogen in HNO_3 .

- A** -1
- B** +1
- C** +3
- D** +5

QUESTION 2

Which type of reaction occurs when two amino acids join to form a dipeptide?

- A** hydrolysis reaction forming a peptide bond
- B** condensation reaction forming a peptide bond
- C** condensation reaction forming an ester bond
- D** hydrolysis reaction forming a glycosidic bond

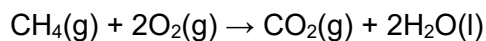
QUESTION 3

Determine the pOH of a 0.0050 M KOH solution at 25°C.

- A** 2.30
- B** 5.00
- C** 9.00
- D** 11.7

QUESTION 4

1.0 mol of methane (CH_4) and 1.0 mol of oxygen (O_2) are reacted according to the equation:



What is the theoretical maximum amount of water (H_2O) that can be produced?

- A 0.5 mol
- B 1.0 mol
- C 2.0 mol
- D 4.0 mol

QUESTION 5

Which pair of compounds have the same molecular formula but different structural formulas (i.e. are structural isomers)?

- A Butan-2-ol and methylpropan-2-ol
- B Ethanol and ethanal
- C Propanoic acid and propan-1-ol
- D Butanal and butanone

QUESTION 6

Which of the following statements best describes the physical properties of straight-chain alkanes with 1–6 carbon atoms?

- A They have low melting points and are soluble in non-polar solvents.
- B They have high melting points and are soluble in polar solvents.
- C They have low melting points and are insoluble in non-polar solvents.
- D They have high melting points and are soluble in water.

QUESTION 7

Which of the following is an oxidation reaction?

- A** Ethene reacting with hydrogen to form ethane
- B** Propanol converting to propanoic acid
- C** Butanoic acid reacting with ethanol to form an ester
- D** An alkyl halide reacting with ammonia to form a primary amine

QUESTION 8

According to collision theory, which change will always increase the rate of a chemical reaction?

- A** Increasing the temperature because this increases the percentage of particles with kinetic energy greater than the activation energy
- B** Increasing the pressure because this increases the number of collisions with energy greater than the activation energy
- C** Adding a catalyst because this lowers the temperature required for the reaction to proceed
- D** Decreasing the concentration because this decreases the total number of effective collisions

QUESTION 9

Which of the following is an acid–base reaction?

- A** $\text{NH}_3(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{NH}_4\text{Cl}(\text{aq})$
- B** $\text{Na}_2\text{CO}_3(\text{aq}) + \text{CaCl}_2(\text{aq}) \rightarrow \text{CaCO}_3(\text{s}) + 2\text{NaCl}(\text{aq})$
- C** $\text{Zn}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{ZnCl}_2(\text{aq}) + \text{H}_2(\text{g})$
- D** $\text{CuSO}_4(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{Cu}(\text{OH})_2(\text{s}) + \text{Na}_2\text{SO}_4(\text{aq})$

QUESTION 10

An electrolytic cell contains a 1.0 M copper(II) chloride (CuCl_2) solution. What products would form at the electrodes?

	Anode	Cathode
A	$\text{H}_2(\text{g})$	$\text{Cu}(\text{s})$
B	$\text{Cl}_2(\text{g})$	$\text{Cu}(\text{s})$
C	$\text{Cl}_2(\text{g})$	$\text{H}_2(\text{g})$
D	$\text{O}_2(\text{g})$	$\text{Cu}(\text{s})$

QUESTION 11

Which of the following molecules exhibits geometric (*cis/trans*) isomerism?

- A 1,2-Dichloroethane
- B 1,1-Dibromoethene
- C But-1-ene
- D 2-Chlorobut-2-ene

QUESTION 12

A 0.050 M solution of KOH can be described as a

- A dilute solution of a weak base.
- B dilute solution of a strong base.
- C concentrated solution of a weak base.
- D concentrated solution of a strong base.

QUESTION 13

The condensed structural formula of a compound is $\text{CH}_3\text{COOCH}_2\text{CH}_3$. What is the IUPAC name of the compound?

- A Ethyl methanoate
- B Methyl ethanoate
- C Ethyl ethanoate
- D Methyl methanoate

QUESTION 14

The polymer nylon 66 is a polyamide formed by the condensation reaction of which two monomers?

- A Amino acids
- B Aminoethanol and ethanoic acid
- C Hexane-1,6-diamine and hexane-1,6-dioic acid
- D Glucose monomers

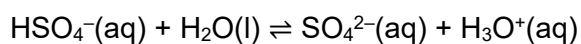
QUESTION 15

Calculate the concentration of OH^- ions in solution if the $[\text{H}^+] = 3.2 \times 10^{-5} \text{ M}$.

- A $3.1 \times 10^{-10} \text{ M}$
- B $3.2 \times 10^{-9} \text{ M}$
- C $3.1 \times 10^{-8} \text{ M}$
- D $3.2 \times 10^{-10} \text{ M}$

QUESTION 16

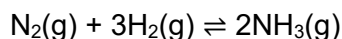
Identify the species that acts as the conjugate base of the $\text{HSO}_4^-(\text{aq})$ ion in the following equilibrium reaction.



- A $\text{HSO}_4^-(\text{aq})$
- B $\text{H}_2\text{O}(\text{l})$
- C $\text{SO}_4^{2-}(\text{aq})$
- D $\text{H}_3\text{O}^+(\text{aq})$

QUESTION 17

For the equilibrium reaction:



which of the following changes would result in the formation of the most $\text{NH}_3(\text{g})$ at equilibrium?

- A Increasing the temperature
- B Increasing the pressure
- C Adding more $\text{H}_2(\text{g})$
- D Adding a catalyst

QUESTION 18

Identify the compound with the lowest volatility, based on the intermolecular forces between its molecules.

- A Propane (C_3H_8)
- B Propan-1-ol ($\text{C}_3\text{H}_7\text{OH}$)
- C Propanal ($\text{C}_2\text{H}_5\text{CHO}$)
- D Propanoic acid ($\text{C}_2\text{H}_5\text{COOH}$)

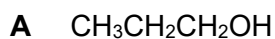
QUESTION 19

Polysaccharides are organic molecules and typically contain

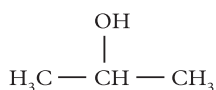
- A ketone or amine groups and hydroxyl groups.
- B hydroxyl groups and glycosidic linkages.
- C aldehyde or ketone groups and hydroxyl groups.
- D carboxyl or aldehyde groups and hydroxyl groups.

QUESTION 20

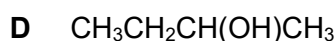
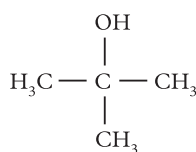
Which of the following structural formulas represents a tertiary alcohol?



B



C



Section 2: Short response

Instructions

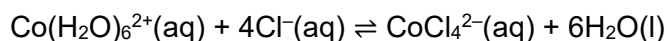
- Write in black or blue pen.
- If you need more space for a response, use the additional pages at the back of the book.
 - On the additional pages, write the question number you are responding to.
 - Cancel any incorrect response by ruling a single diagonal line through your work.
 - Write the page number of your alternative/additional response, i.e. “See page ...”
 - If you do not do this, your original response will be marked.
- This section has **6 questions** and is worth **31 marks**.

QUESTION 21 (1 mark)

Explain why a chemical equilibrium can only be established in a closed system. [1 mark]

QUESTION 22 (3 marks)

5 mL of a solution containing $\text{Co}^{2+}(\text{aq})$ ions (pink) is mixed with 6 M $\text{Cl}^{-}(\text{aq})$ ions (from concentrated HCl, colourless) in a large test tube at room temperature. The following equilibrium is established.



pink

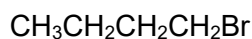
blue

- a** When more concentrated HCl is added, the solution turns a deeper blue. Explain this colour change in terms of equilibrium position. [1 mark]

- b The test tube is then placed in an ice–water bath, and the solution becomes pink again. Deduce whether the reaction is endothermic or exothermic, and justify your answer using Le Châtelier’s principle. [2 marks]

QUESTION 23 (4 marks)

A structural isomer of C_4H_9Br is shown below.



- a Explain why this compound is classified as a saturated compound. [1 mark]
- b Identify whether this compound is a primary, secondary or tertiary haloalkane. [1 mark]
- c Draw the condensed structural formula of another isomer of C_4H_9Br that is a secondary haloalkane and give its IUPAC name. [2 marks]

QUESTION 24 (11 marks)

Organic chemistry requires the careful classification of compounds based on their molecular formula, structure and functional groups. The compound C_4H_9Cl and the alcohol propan-1-ol (C_3H_8O) demonstrate key principles of isomerism and chemical reactivity.

- a Describe and explain structural isomerism in substituted alkanes. Choose a hydrocarbon derivative with the formula C_4H_9Cl and draw appropriate structural diagrams in the space below to support your answer. [3 marks]

- b** Draw a structural isomer of propan-1-ol that belongs to a different functional group class. *[1 mark]*

- c** Identify an appropriate chemical test to distinguish between the two functional group classes represented by propan-1-ol and the isomer you drew in part b. Explain how you could conclude which is which. *[2 marks]*

- d Draw and provide the IUPAC names of the possible organic products formed by reaction of the compound prop-1-ene with hydrogen bromide (HBr). [2 marks]

- c Explain how the major product formed in part b could be used to synthesise a ketone. Your answer must include the name of the final product and the necessary reagents and conditions. [3 marks]

QUESTION 25 (9 marks)

Electrochemical cells, such as galvanic cells, operate by exploiting spontaneous redox reactions. Understanding the fundamental components of these cells, their standard potential, and the principles that govern cell potential under non-standard conditions is essential for the study of electrochemistry.

- a Identify and describe the roles of the three key components of an electrochemical cell (excluding the wires and voltmeter) and explain the purpose of the salt bridge. [4 marks]

- b** A galvanic cell is set up using a Fe^{2+}/Fe half-cell, both under standard conditions. Determine the balanced net ionic equation for the spontaneous cell reaction and calculate the standard cell potential (E°_{cell}). (Assume $E^\circ_{\text{Fe}^{2+}/\text{Fe}} = -0.44 \text{ V}$ and $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = +0.34 \text{ V}$). [2 marks]

- c** For the Fe^{2+}/Fe half-cell used in part b, a small amount of solid sodium hydroxide (NaOH) is added. Using Le Chatelier's principle, predict and explain the effect this addition will have on the half-cell potential (E°_{half}). [3 marks]

QUESTION 26 (3 marks)

A student performs a titration between 0.1 M ethanoic acid (CH_3COOH) and 0.1 M sodium hydroxide (NaOH). Identify the initial pH range, the pH at the equivalence point, and the most suitable indicator for this titration. Justify the indicator choice.

END OF PAPER

